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Gavin Higgins

Education

University of Washington, Seattle — Seattle, WA

Bachelor of Science - Mechanical Engineering, 3.60 GPA — *Sep 2025 - Jun 2029*

Experience

Hero Robot Lead — Seattle, WA

Advanced Robotics at the University of Washington — *October 2025 - Present*

• Engineered a low-backlash yaw mechanism for the 'Hero' robot. Used nTop generative design and Solidworks FEA to increase impact resistance and overall stiffness. Robot placed 3rd in international competition.

• Replaced faulty drive pulley mounting with zero backlash components. Introduced serviceable sliding belt tensioner to ensure ease of control throughout the robot's lifetime. Total robot weight reduced by 28%.

• Leading a team of 5 to fully redevelop the robot for the 2027 season, aiming to reduce weight, increase electronics accessibility, increase shot accuracy, and place 1st in competition.

Robot Lead — Scottsdale, AZ

High School FRC Robotics Competition Team — *August 2022 - May 2025*

• Directed a team of 10 rookies to design and build a 125 lb competition robot, using mainly in-house manufacturing during a 6 week timeline. Created the best-performing robot in 15-year team history.

• Eliminated previous robots' constant electrical and mechanical failures, and modernized style of building robots to guarantee competitiveness. Reshaped internal curriculum and removed previous leadership team issues.

Projects

Omni Wheel Suspension Module & Spring Optimization

Advanced Robotics at the University of Washington — *September - October 2026*

• Designed a suspension and drive module for use on two ARUW 2027 robots. Sturdy construction of 0.25" 6061 and 4mm carbon fiber plates, combined with dual MGN9H rails for motion, correct faults of previous year's defective modules.

• Validated structure via ANSYS, handles drops from 200% higher and lateral impacts of 560 N, 4x more than 2026.

• Wrote MATLAB script to optimize spring and damper constants with drop test and bumpy surface stability constraints, to ensure core robot CV targeting and firing system is precise throughout unpredictable field obstacles.

PID Wheel-Legged Balancing Robot

Personal Project — *September 2026 - In Progress*

• ESP32-based bipedal wheel-legged robot built to learn control theory. Currently balances in place.

• Goals: perform jumps and drops, remote control, fall recovery, LQR implementation.

Skills

Technical Skills: Solidworks, ANSYS, Onshape, nTop, Manual and CNC Machining

Relevant Coursework: Differential Equations, Linear Algebra, Engineering Statics
In Progress Coursework: Kinematics and Dynamics, Fundamentals of Material Science, Electromagnetism